A Walk-Forward Ridge Forecaster for Ten ETFs

Machine-learning strategies for investment research, tested honestly: real data, next-open fills, overfitting statistics, and no evidence of alpha.

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ABSTRACT A ridge regression forecasts 21-session returns for ten liquid ETFs from trend, momentum, volatility and liquidity features. A long-only optimiser turns the forecasts into weights, and the system is tested walk-forward on real daily data from 2009 to 2026 with a 21-session embargo, next-open fills and 5 bps costs. It returned 7.05% a year at 10.2% volatility, with a Sharpe ratio of 0.72 and a −23.0% maximum drawdown. SPY returned 14.7% and an equal-weight basket 9.3%. Alpha against SPY is not significant ($t = 0.49$), and macro factors explain 70% of its variance. The overfitting statistics are benign (PBO 0.21) but cover only ten ridge penalties. The honest reading is a diversified risk-premia allocation, not a source of edge.

01

Summary

VERDICT

The model passes its overfitting checks and fails to show alpha. It earns less than SPY and less than its own equal-weight baseline. What it delivers is lower volatility and a shallower drawdown, which a static mix of SPY and cash would also deliver. That is a trustworthy negative result about edge, and a positive one about the research framework.

7.05%	0.72	t = 0.49	0.21
Annualised return net of 5 bps, 2009–2026 (SPY 14.7%, equal-weight 9.3%)	Sharpe ratio, arithmetic convention (0.69 as return ÷ volatility)	Alpha vs SPY, 0.81% a year (p = 0.62): not significant	Probability of backtest overfitting across 10 ridge penalties

KEY FINDINGS

1. **No evidence of alpha.** Against SPY the strategy has a beta of 0.43 and an annual alpha of 0.81% with t = 0.49. Five macro factors explain 70% of its daily variance and leave an alpha of −0.78% (t = −0.58).
2. **Lower return, lower risk, nothing more.** Annual return was 7.05% against 14.7% for SPY and 9.3% for an equal-weight basket of the same ETFs. Upside and downside capture against SPY are almost identical (0.48 and 0.49), the signature of a scaled-down beta, not of skill in timing.
3. **Overfitting statistics look clean, within a narrow search.** The deflated Sharpe ratio is 0.998 and PBO is 0.21. Both deflate only over ten ridge penalties. The universe, horizon, features and risk limits were chosen with the full history in view, and no statistic here accounts for that.
4. **The plumbing is sound where leakage usually hides.** The test purges overlapping labels with a 21-session embargo, fills at the next open and charges costs on every trade. It also has gaps: it is long-only, uses flat costs, and saves no forecast-skill metric such as an information coefficient.

The repository's own design notes reach the same conclusion. They describe the system as a research *framework* and, in their own words, “not an alpha engine”. This report documents why that framing is the right one, and what evidence would be needed to change it.

02

Question, universe and data

The repository states no formal hypothesis. Its implicit question is whether trailing trend, momentum, volatility and liquidity features predict the next month's returns across a cross-asset ETF universe, and whether a long-only, risk-limited portfolio built from those forecasts earns a better risk-adjusted return after costs than holding the benchmark.

Table 1. The ten-ETF universe (“mvp-etf-v1”), with buy-and-hold return over the test period and the model's final forecast and target weight.

ETF	Asset class	Buy-and-hold, Jan 2009 – Jun 2026	21-session forecast, 12 Jun 2026	Target weight
QQQ	US equity	+2,594%	1.15%	11.6%
SPY	US equity	+992%	0.91%	16.1%
IWM	US equity	+634%	0.91%	9.8%
VNQ	Real estate	+470%	0.50%	8.2%
GLD	Gold	+358%	0.57%	5.6%
EFA	International equity	+291%	0.66%	9.2%
EEM	International equity	+266%	1.16%	8.2%
DBC	Commodities	+52%	0.57%	8.4%
IEF	Rates	+46%	0.21%	9.9%
TLT	Rates	+27%	0.30%	8.1%

Buy-and-hold is stored in the dashboard under the key per_symbol_contribution, but the code computes it as each ETF's simple close-to-close return over the backtest, not the strategy's P&L by holding. Cash at the final rebalance: 5.0%.

Data. Daily bars come from Yahoo Finance via yfinance with split and dividend adjustment (total-return adjusted), from 3 January 2007 to 12 June 2026. That is 4,892 sessions per ETF and 48,920 rows, complete against the NYSE calendar with no excluded sessions. The snapshot is frozen and hashed. The out-of-sample evaluation runs from 5 January 2009 to 12 June 2026, 4,387 sessions or 17.4 years. Its first eight months are flat because the model needs enough labelled history before it trades, so the first non-zero month is September 2009.

All ten ETFs traded throughout the period, so delisting bias is small. The universe itself, though, was picked with hindsight: these are liquid funds that survived and stayed relevant. The repository lists point-in-time universe handling as unfinished work.

03

Method

Features, label and model

Six features are computed from each ETF's own history up to the decision date: simple returns over 21, 63 and 126 sessions; 21-session realised volatility (annualised standard deviation of log returns); distance from the 100-session moving average; and 20-session average dollar volume. The label is the forward return over the next 21 sessions. Features are z-scored across the ten ETFs on each date, and one ridge regression with penalty $\alpha = 1.0$ is fitted on all ETFs pooled.

Walk-forward validation

The training window expands. The model needs at least 504 distinct training sessions and is refitted every 21 sessions. A training row is usable only if its label ends before the embargo, which starts 21 sessions before the decision date. Because every label spans 21 sessions, this rule purges any label that overlaps the forecast horizon; the code enforces it as a hard check. The decision is made at the close of session t and filled at the open of session $t + 1$, and positions are marked at the close.

Portfolio rule and costs

Only ETFs with a positive forecast are eligible. Desired weights are proportional to forecast ÷ trailing volatility and are scaled so the portfolio is 95% invested. A convex solver then finds the nearest weights that satisfy four limits.

$$\min \|w - w^{*}\|^2 \quad s.t. \quad w \geq 0, \quad w_i \leq 0.25, \quad \Sigma w_i \leq 0.95, \quad w' \, diag(\sigma^2) \, w \leq 0.15^2 \tag{1}$$

The volatility limit treats the ETFs as uncorrelated, a simplification the code documents. Every buy and sell pays a flat 5 bps of notional, and results are also reported at 2 and 10 bps. Two baselines run on the same data. One is SPY bought and held. The other is an equal-weight basket that holds 9.5% in each ETF, is rebalanced every session and pays the same costs.

> LONG-ONLY
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> The README describes “systematic long/short trading”, but the optimiser forces w ≥ 0 and the project notes defer shorting. Every result in this report is long-only.

04

Results

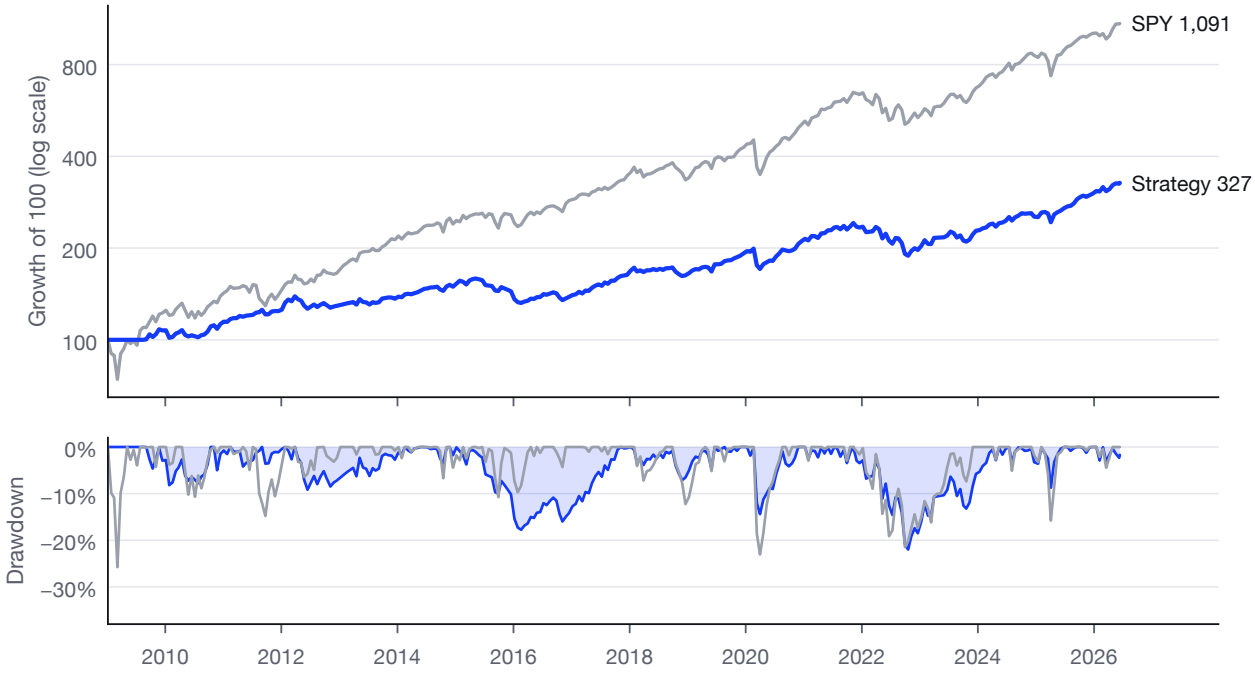


Figure 1. Growth of 100 on a log scale (top) and drawdown (bottom) for the strategy, net of 5 bps (blue), and SPY buy-and-hold (grey). Both series are stored at 315 sample dates, so troughs between samples are missed. The strategy's daily maximum drawdown is −23.0%.

Source: mltrade web/public/data/dashboard.json (performance.equity_curve, performance.drawdown).

Table 2. Out-of-sample performance, 5 Jan 2009 – 12 Jun 2026 (4,387 sessions).

Measure	Strategy (5 bps)	SPY	Equal-weight
Annualised return	7.05%	14.71%	9.33%
Growth of $1m	$3.27m	$10.91m	—
Annualised volatility	10.19%	—	—
Sharpe (arithmetic / return ÷ volatility)	0.720 / 0.692	—	—
Sortino / Calmar	1.00 / 0.31	—	—
Maximum drawdown	−22.97%	—	—
Longest drawdown / time to recover	637 / 396 sessions	—	—
Daily skew / excess kurtosis	−0.52 / 8.0	—	—
Best / worst day	+6.04% / −5.39%	—	—
Daily VaR / CVaR, 95%	1.01% / 1.57%	—	—

SPY's annualised return is derived here from its stored equity curve with the same formula that reproduces the strategy's 7.05%. The dashboard saves only an annualised return for the equal-weight baseline. A dash means no comparable figure was saved.

Over 17.4 years the strategy turned $1m into $3.27m, while SPY turned it into $10.91m. The return gap, 7.7 percentage points a year, is far larger than anything the volatility reduction buys back. A blend of about 43% SPY and 57% cash would match the strategy's beta. The strategy's 0.81% annual alpha over that blend is statistically indistinguishable from zero. The return distribution has a fat left tail: skew is −0.52, excess kurtosis is 8.0, and the worst day lost 5.39%.

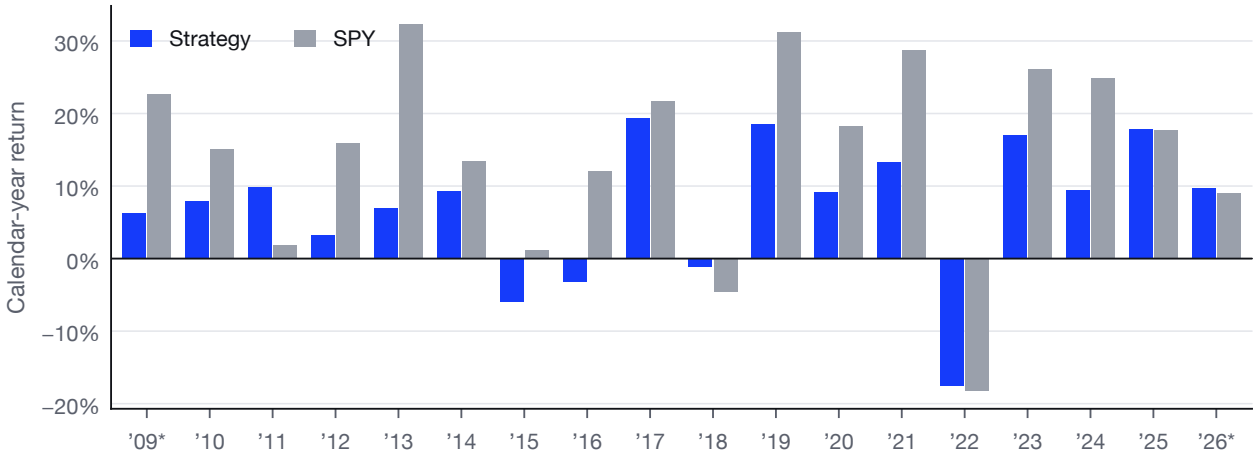


Figure 2. Calendar-year returns, strategy (blue) and SPY (grey). * 2009 includes eight flat warm-up months; 2026 runs to 12 June.
Source: dashboard.json (performance.yearly).

The strategy beat SPY in 5 of 18 calendar years: 2011, 2018, 2022, 2025 and 2026 to date. The margin was large in 2011 and 2018 (7.9 and 3.5 points) and under one point in the other three. It lost money in four years (2015, 2016, 2018 and 2022). Its worst year, 2022, cost 17.5% against 18.2% for SPY: in the one year when stocks and bonds fell together, diversification across the universe protected almost nothing. That reading is an inference from the yearly figures and the portfolio's duration exposure (Section 05), not a separately tested result.

05

Robustness

Stability across time

Fifteen of 18 windows have a positive Sharpe ratio, and the median is 0.98. The spread is wide, from −0.96 in the window starting January 2022 to 3.61 in 2017, so no single year is representative. By regime, the strategy was positive through 2020 (+9.2%) and 2024 (+9.4%) and negative through 2015–16 and 2022. The 2008 crisis falls before the evaluation window.

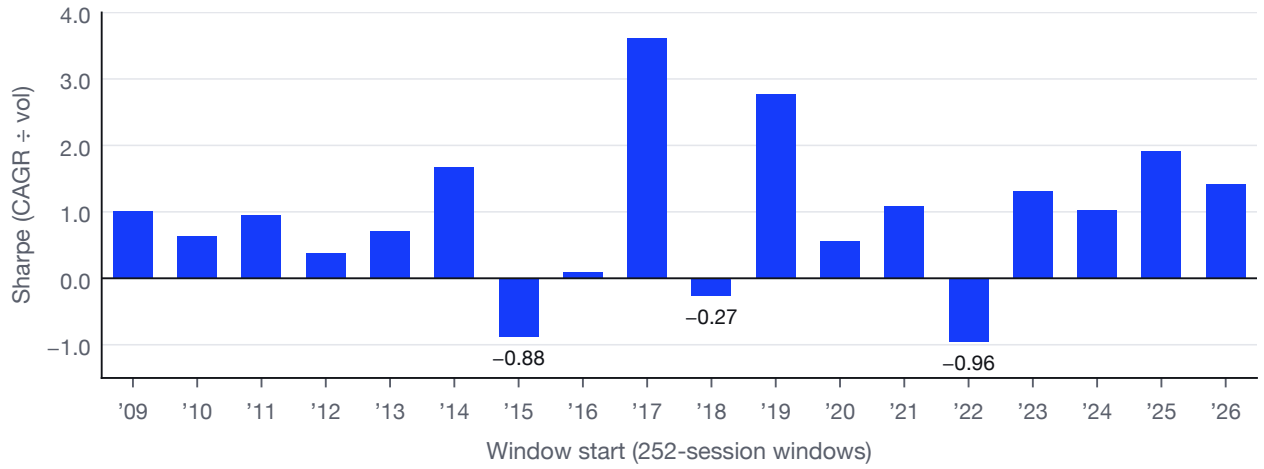


Figure 3. Sharpe ratio in 17 consecutive 252-session evaluation windows plus a partial 2026 window, computed as annualised return ÷ volatility. Three windows are negative: those starting in 2015 (–0.88), 2018 (–0.27) and 2022 (–0.96).

Source: dashboard.json (performance.evaluation_windows).

Transaction costs

Table 3. Cost sensitivity. The trading decisions are identical at each cost level; only the cost charged changes.

Cost per side	Annualised return	Sharpe (return ÷ volatility)	Maximum drawdown
2 bps	7.26%	0.712	–22.83%
5 bps (headline)	7.05%	0.692	–22.97%
10 bps	6.70%	0.658	–23.19%

Each extra basis point of cost removes about 0.07 percentage points of annual return. That fits the reported average turnover of 2.6% of equity per session, about 6.5 times equity a year. Total costs at 5 bps came to $93,362, which implies roughly $187m of traded notional. Costs are not what holds the strategy back. Even at 2 bps it trails SPY by more than seven points a year. The sweep is narrow, though, and the model charges a flat rate whatever the ETF's liquidity, with no spread, impact or capacity analysis.

> TWO SHARPE RATIOS
>
> The headline 0.720 and the 5 bps cost-table 0.692 come from the same returns but use different formulas. The headline divides the arithmetic mean daily return by its standard deviation and scales by √252. The cost table divides the geometric annual return by annual volatility. The gap is the usual σ²/2 drag: ln(1.0705) + 0.1019²/2 = 7.33%, which matches the 7.33% annual arithmetic mean implied by the headline. The repository does not reconcile the two in its output; this report quotes both.

What the returns are made of

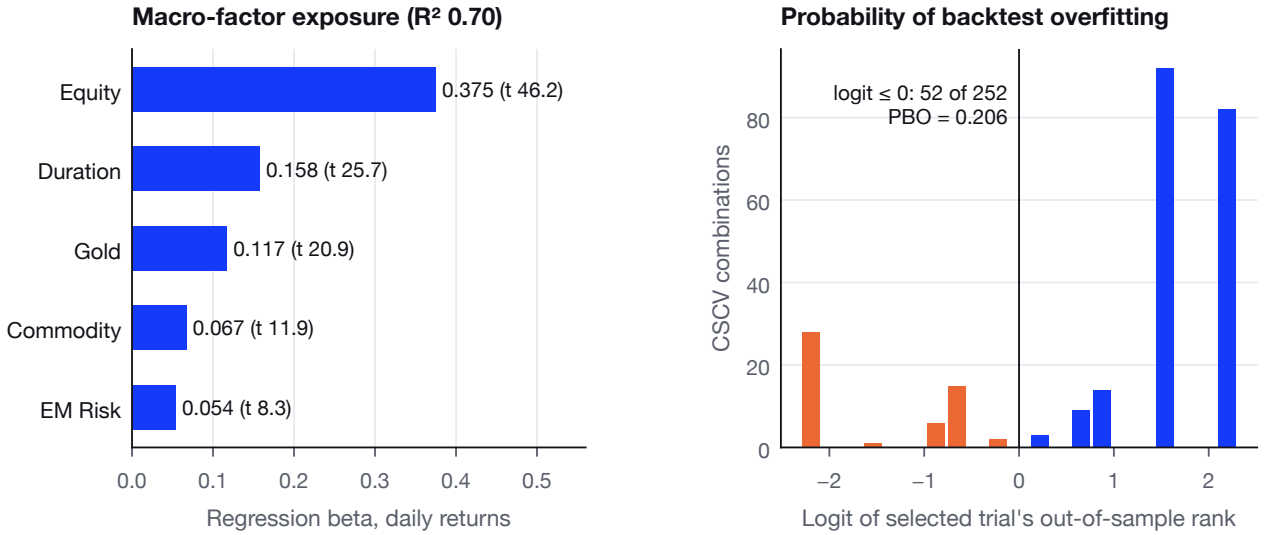


Figure 4. Left: daily-return regression on five ETF-proxy macro factors (Equity = SPY, Duration = TLT, Gold = GLD, Commodity = DBC, EM Risk = EEM), betas with t-statistics. Right: combinatorially symmetric cross-validation over ten ridge penalties. Orange bars are combinations where the in-sample winner ranked in the bottom half out of sample.

Source: dashboard.json (attribution, overfitting.logit_histogram).

The factor regression explains 70% of the strategy's daily variance. Every loading is positive and precisely estimated: equity 0.375 (t = 46.2), duration 0.158, gold 0.117, commodity 0.067 and EM risk 0.054. What remains is an alpha of −0.78% a year (t = −0.58). That is the profile of a largely static allocation to well-known risk premia. One likely reason, which this report does not test, is the portfolio rule: whenever any forecast is positive the optimiser scales to 95% invested, and at the final rebalance all ten forecasts were positive (Table 1).

Validation scorecard

Table 4. The validation checks a promotion decision would need, and what the repository's saved run shows.

Check	Result	Status
Walk-forward with purge/embargo	Expanding window, 21-session embargo, refit every 21 sessions	**Done**
Realistic fills and costs	Next-open fills; 2 / 5 / 10 bps flat	**Partly**
Alpha vs benchmark	0.81% a year, t = 0.49, p = 0.62	**Not significant**
Deflated Sharpe ratio	0.998 over 10 trials	**Passes, narrow**
Probabilistic Sharpe ratio vs 0	0.998	**Passes**
Probability of backtest overfitting	0.206 (52 of 252 combinations; 10 splits)	**Passes, narrow**
Regimes (2015–16, 2020, 2022, 2024)	Positive in 2020 and 2024; negative in 2015–16 and 2022	**Mixed**
Bootstrap confidence interval on Sharpe	Not computed	**Missing**

Check	Result	Status
Forecast skill (IC, hit rate of forecasts)	Not computed; the stored “hit rate” 0.533 is the share of positive days	Missing
Untouched holdout	None; design choices saw the full sample	Missing
Capacity	Not analysed	Missing

06

Limitations

- **Design hindsight.** The universe, the 21-session horizon, the features, the 25% cap and the 15% volatility target were fixed after seeing the whole history. The deflation statistics cover only the ridge penalty, so they understate the real search.
- **No measure of forecast quality.** Without an information coefficient it is impossible to tell whether the ridge forecasts carry any cross-sectional information, or whether the returns come entirely from being invested.
- **Simplified execution.** The model uses a flat cost rate, with no bid-ask spread by instrument, no market impact and no taxes. Live trading is disabled, and the order preview runs against a simulated broker. On the final date, two of 17 risk gates blocked the rebalance because of a notional limit.
- **Simplified risk model.** The volatility limit uses a diagonal covariance, so it ignores the equity–bond correlation that mattered most in 2022.
- **Vendor data.** Yahoo's adjusted prices are revised retroactively and are not a point-in-time record.
- **Unused experiment platform.** The experiment and Optuna tooling has no saved runs, and its configuration files point at a synthetic fixture. A dashboard mock-up in the repository shows that synthetic run (Sharpe 2.25); none of its numbers are used here.

What would change the verdict

An edge claim would need four things the repository does not yet have. The first is a saved information coefficient for the forecasts, by date and by ETF. The second is a bootstrap interval on Sharpe and on alpha. The third is a deflation that counts every configuration actually tried. The fourth is a holdout period that no design decision has touched. Until then, the right comparison for this system is a volatility-matched SPY-and-cash mix, and against that mix it shows no difference.

A

Data and reproducibility

REPOSITORY	ajaiupadhyaya/mltrade, commit 820ccdd, 16 Jun 2026. The working tree was clean when read.
RESULTS FILE	web/public/data/dashboard.json (schema 2, generated 13 Jun 2026, data mode “real-frozen-snapshot”, synthetic = false). It holds the only saved real-data backtest; no other run records exist.
MARKET DATA	data/snapshots/real/daily_bars_2026-06-13: 48,920 rows, 10 ETFs, 3 Jan 2007 – 12 Jun 2026, content SHA-256 253a662c…4b45. Source: yfinance with auto-adjust.

MODEL	ridge-trend-v1 on features trend-momentum-v1. Universe mvp-etf-v1. Reference equity $1,000,000.
CODE	features/definitions.py, models/walk_forward.py, portfolio/optimizer.py, backtest/engine.py, analytics/ (returns, attribution, overfitting).
VERIFICATION	The project notes report 430 passing tests (15 Jun 2026); they were not re-run for this report. The backtest was not re-run either.
THIS REPORT	All four figures were redrawn from dashboard.json. Derived here from the same file: SPY's annualised return, which uses the formula that reproduces the strategy's 7.05% (checked); the Sharpe reconciliation; PBO recomputed from the logit histogram (checked); the number of positive windows; the year-by-year comparisons; the cost drag per basis point; and annual turnover.

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